KIRAN SAPKOTA

Machine Learning Engineer · Data Scientist · AI Engineer

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PROFESSIONAL SUMMARY

Results-driven Machine Learning Engineer and Computer Systems Engineering graduate (University of Sunderland, 2025) with proven hands-on experience building end-to-end ML and NLP pipelines. Skilled in designing, training, and evaluating deep learning models using Python, PyTorch, and Scikit-learn. Demonstrated ability to deliver production-quality solutions in agile, international team environments through the Omdena Global AI Innovation Challenge. Adept at translating complex data into actionable insights and deployable AI systems that generate measurable real-world impact.

CORE COMPETENCIES

ML & AI	Supervised & Unsupervised Learning, Neural Networks, Attention Mechanisms, Model Optimization, Performance Evaluation
NLP / DL	BiLSTM, BERT, Hugging Face Transformers, TF-IDF Vectorization, Text Classification, Sequence Modelling
Frameworks	PyTorch, TensorFlow, Scikit-learn, Keras, NLTK
Data Science	EDA, Feature Engineering, Statistical Analysis, Data Preprocessing, Data Visualisation
Programming	Python (advanced), SQL, Bash / Shell scripting
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Dev & Tools	Git, GitHub, Linux, Agile / Scrum, Networking & IT Infrastructure
Soft Skills	Analytical Thinking, Problem-Solving, Cross-cultural Teamwork, Communication, Adaptability

WORK EXPERIENCE

Junior Machine Learning Engineer (Collaborator)

Jul 2025 – Sep 2025

Omdena – Global AI Innovation Challenge

- Collaborated on an 8-week global AI project with an international, multidisciplinary team following agile workflows to deliver ML solutions for real-world problems.
- Executed end-to-end data pipelines: collection, cleaning, preprocessing, and exploratory data analysis (EDA) to prepare high-quality datasets for model training.
- Contributed to model development, hyperparameter tuning, and systematic performance evaluation, resulting in measurable accuracy improvements.
- Communicated findings clearly across time zones, maintaining documentation and participating in sprint reviews with stakeholders.
- Awarded the Omdena Junior Machine Learning Engineer Badge for exemplary contribution to the project deliverables.

FEATURED PROJECTS

Fake News Detection System

View on GitHub ↗

Tech: *Python, Scikit-learn, NLTK, PyTorch, TF-IDF, Logistic Regression, Random Forest, BiLSTM, BERT*

- Built a complete NLP classification pipeline on GossipCop & PolitiFact datasets, including tokenisation, stopword removal, stemming, and TF-IDF feature extraction with NLTK.
- Fine-tuned Logistic Regression and Random Forest classifiers achieving approximately 98% test accuracy through rigorous cross-validation and hyperparameter optimisation.
- Benchmarked deep learning approaches (BiLSTM, BERT) and produced detailed visualisations of precision, recall, F1-score, and ROC-AUC to compare model performance.

Resume Classification System

View on GitHub ↗

Tech: *Python, PyTorch, BiLSTM, Attention Mechanism, TF-IDF, XGBoost, Scikit-learn, Resume-Atlas Dataset*

- Designed and trained a PyTorch BiLSTM model with a custom attention mechanism on the Resume-Atlas dataset, achieving 86% classification accuracy — a 4-point improvement over the XGBoost baseline (82%).
- Implemented a complete preprocessing pipeline including text normalisation, TF-IDF vectorisation, and multi-class label encoding for 25+ job role categories.
- Delivered top-3 job role predictions with confidence scoring, making the system directly applicable to intelligent recruitment automation workflows.

EDUCATION

BSc (Hons) Computer Systems Engineering

2022 – 2025

University of Sunderland, UK

- Relevant modules: Machine Learning, Data Structures & Algorithms, Computer Networks, Database Systems, Software Engineering, Operating Systems.

Higher Secondary Education (Science)

2019 – 2021

New Oasis Academy

Secondary Education Examination (SEE)

2018

Corona Academy

CERTIFICATIONS & AWARDS

Foundational Data Science Course

Omdena Academy · Nov 2025

- Covered Python for data science, exploratory data analysis (EDA), core ML algorithms, data visualisation, and a capstone web application project integrating a deployed ML model.

Junior Machine Learning Engineer Badge

Omdena · Sep 2025

- Recognised for successful contributions to an 8-week global AI project involving real-world ML model development and international collaboration.

LANGUAGES

English – Professional Working Proficiency

Nepali – Native / Full Professional Proficiency